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Bihar Engineering University, Patna

B.Tech 5th Semester Exam., 2025

Code: 105501 ***Time: 03 Hours***

Semester: V ***Subject: ARTIFICIAL INTELLIGENCE*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) The Turing Test was proposed by
(i) John McCarthy
(ii) Alan Turing
(iii) Marvin Minsky
(iv) Herbert Simon

(b) Weak AI is also known as
(i) Narrow AI
(ii) General AI
(iii) Strong AI
(iv) Super AI

(c) Breadth First Search is an example of
(i) Informed Search
(ii) Uninformed Search
(iii) Heuristic Search
(iv) Local Search

(d) Which algorithm is optimal and complete?
(i) Depth First Search
(ii) Hill Climbing
(iii) A* Search
(iv) Simulated Annealing

(e) Heuristic function h(n) is used in
(i) Random Search
(ii) Brute Force Search
(iii) Uninformed Search
(iv) Informed Search

(f) AO* algorithm is used for solving
(i) AND-OR graphs
(ii) Game trees
(iii) Decision trees
(iv) Neural networks

(g) Constraint Satisfaction Problems are usually solved using
(i) Backtracking
(ii) Breadth First Search
(iii) A* Search
(iv) Alpha-Beta Pruning

(h) Which of the following is a stochastic search method?
(i) Breadth First Search
(ii) Particle Swarm Optimization
(iii) Depth First Search
(iv) A* Search

(i) Bayesian Networks are used to handle
(i) Certain knowledge
(ii) Uncertain knowledge
(iii) Deterministic knowledge
(iv) Syntax only

(j) Neural Networks are inspired by
(i) Genetic evolution
(ii) Human brain
(iii) Particle movement
(iv) Graph theory

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Q.2 **[7+7=14]**

(a) Explain what you understand by Artificial Intelligence. Describe the Turing Test in detail and discuss its significance in the field of AI.

(b) Explain the concept of Rational Agents. How do they differ from Reflex Agents?

Q.3 **[7+7=14]**

(a) Evaluate the impact of AI on employment and society.

(b) Explain Breadth First Search and Depth First Search with examples.

Q.4 **[7+7=14]**

(a) Describe the role of heuristics in A* algorithm.

(b) "A* algorithm is the best search technique." Do you agree? Justify your answer with reasoning.

Q.5 **[7+7=14]**

(a) Describe Alpha-Beta Pruning. Why is it used in game playing?

(b) Describe the working of simulated annealing with suitable example.

Q.6 **[7+7=14]**

(a) Explain AO* algorithm and how it differs from A* algorithm.

(b) Discuss the major challenges in developing a practical Robotics system using AI.

Q.7 **[7+7=14]**

(a) Explain Bayesian Networks and their role in handling uncertainty.

(b) Convert the following statements into First Order Logic: "All men are mortal. Socrates is a man."

Q.8 **[7+7=14]**

(a) Explain the difference between Supervised, Unsupervised, and Semi-Supervised Learning with examples.

(b) Describe the architecture of a simple Artificial Neural Network. Explain the role of activation functions.

Q.9 Write short note on any two of the following: **[7 x 2 = 14]**

(a) Natural Language Processing

(b) Expert Systems

(c) Deep Learning

(d) Computer Vision

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